

Autonomous Blocking via Trajectory Switching for F1TENTH Racing

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Abstract—This paper presents a robust autonomous racing framework for defensive maneuvering in competitive environments, using the F1TENTH platform. Unlike standard single-path tracking, our approach uses a Multi-Lane Pure Pursuit controller that can switch between pre-calculated global racelines (inner, middle, and outer) in real time. By integrating YOLOv11-based opponent detection with a hybrid distance estimation to reliably track opponent position and range, a reactive finite-state manager dynamically selects the optimal raceline to defend against overtaking maneuvers while maintaining near-optimal racing velocity when no car is detected. Experimental results demonstrate the effectiveness of path switching in strategically blocking opponent vehicles.

Index Terms—Autonomous Racing, Path Planning, Pure Pursuit, F1TENTH, Defensive Driving, ROS 2.

I. INTRODUCTION

AUTONOMOUS racing is widely used to study motion planning and control methods for high-speed autonomous driving. Early work in this field primarily addressed time-optimal driving, where the objective is to minimize lap time by following a single optimal raceline. [1] [2] However, real racing scenarios involve multiple vehicles, where tactical behavior becomes necessary. In such situations, the fastest path is not always the best choice, as a vehicle must also defend its position against opponents attempting to overtake. Therefore, developing a system that can “block” or defend a lane is essential for winning races.

This paper investigates defensive tactical behavior in autonomous racing on the F1TENTH platform. Conventional Pure Pursuit controllers track fixed trajectories [3], exposing vehicles to predictable overtaking, particularly on straight segments and at corner entry. Multi-line switching addresses this by enabling real-time tactical responses. However, this introduces several challenges: maintaining lateral stability during high-speed transitions and making timely decisions with noisy, latent perception data.

Previous approaches have addressed interaction with other vehicles primarily through reactive obstacle-avoidance strategies, such as potential fields [4] or gap-following methods [5]. While effective for collision avoidance, these techniques are not designed to enforce the structured, strategic lane-blocking behavior required in competitive racing scenarios.

To overcome these limitations, we propose a reactive path-switching framework extending Pure Pursuit to multiple racing lines. Three optimized racelines—Inner, Middle, and Outer—are designed to maximize defensive coverage while remaining dynamically feasible. A high-level finite-state manager selects the appropriate line based on relative opponent position, enabling effective tactical defense.

Success is measured by the system’s ability to minimize the available overtaking window while maintaining stable racing behavior during lane transitions. Experimental evaluation on the F1TENTH platform demonstrates the feasibility and effectiveness of the proposed approach.

The main contributions of this work are as follows:

- 1) **Raceline Synthesis:** We generate three distinct optimized racelines specifically designed to reduce overtaking opportunities in critical passing zones.
- 2) **Multi-Trajectory Pure Pursuit Controller:** We implement a C++ Pure Pursuit controller capable of tracking multiple global racelines and performing stable real-time transitions between them.
- 3) **High-Level Lane Management Logic:** We develop a Python-based tactical manager that uses opponent distance estimates to trigger defensive lane changes.

II. SYSTEM ARCHITECTURE

The proposed framework is implemented using the Robot Operating System (ROS 2) to ensure a modular and real-time communication pipeline. The architecture is divided into four functional layers:

The **Localization Layer** performs vehicle state estimation using a particle filter within a pre-mapped environment. The **Perception Layer** employs an RGB-D camera (ZED2) together with a YOLOv11 model to estimate the relative position of trailing opponent vehicles. The **Tactical Layer**, implemented as a Python-based manager node, evaluates these distance estimates and publishes the selected raceline index. The **Control Layer** is realized as a high-performance C++ Pure Pursuit controller that generates lateral and longitudinal control commands for the active raceline.

III. METHODOLOGY

The technical implementation of the proposed framework focuses on converting perception data into stable control actions. This section describes the raceline optimization process used to generate defensive trajectories (III-A), the vision-based detection and tracking of the opponent vehicle (III-B), the formulation of the multi-lane Pure Pursuit controller (III-C), and the strategy employed for raceline selection (III-D).

A. Raceline Optimization

To support tactical lane changes in the control layer, three distinct global reference trajectories are generated. An occupancy-grid map of the racetrack is constructed via SLAM. Three defensive racelines are synthesized by generating three map variants whose boundaries are selectively shifted in GIMP to bias the feasible corridor toward an inner, centered, or outer band. Each variant is then processed by a global planning pipeline that converts the grid map into a continuous track representation and solves a staged minimum-time optimization, yielding smooth, dynamically feasible, near time-optimal trajectories. The inner and outer configurations are designed such that the resulting racelines leave a lateral margin to the corresponding boundary that is typically on the order of one vehicle width, limiting practical overtaking space adjacent to the defended line.

B. Vision-Based Detection Model (perception layer)

Reliable defensive maneuvering requires accurate real-time estimation of the opponent vehicle's position. Opponent detection is performed using the YOLOv11 architecture, optimized for high-speed inference on embedded hardware. Due to limited labeled training data, a semi-supervised dataset expansion pipeline is employed. An initial model is trained on a small manually labeled dataset of F1TENTH vehicle images and then used to generate "pseudo-labels" from recorded track-driving videos. To improve robustness, data augmentation is applied, including noise injection, random scaling, and image rotation, and images of empty track segments are added to reduce false positives. The model is retrained on the expanded dataset, improving detection performance under varying lighting and track conditions.

After detection, the opponent's distance is estimated for tactical decision-making. At long range, depth values from the ZED2 stereo camera are used, while at close range a geometry-based formulation is applied using the known opponent height H and camera focal length f :

$$d = \frac{f \times H}{h_{\text{pixels}}}, \quad (1)$$

where h_{pixels} is the height of the detected bounding box in pixels. The estimator selects the most reliable measurement based on range.

If multiple detections occur in one frame, only the closest opponent is tracked. The lateral position is determined by comparing the bounding box center with the image center to classify the opponent as left or right. The distance estimation

node publishes the selected distance and lateral position to a ROS topic, providing the Tactical Layer with a consistent estimate of the opponent's relative state.

C. Multi-Lane Pure Pursuit (control layer)

The control layer is implemented as a high-performance C++ node designed to manage three concurrent racing lines (Inner, Middle, and Outer) through a parallelized state-tracking architecture. To eliminate switching latency, the controller maintains a set of `LineData` structures that store waypoints, lookahead bounds, and proportional gains (K_p) for each lane. At each odometry callback, the `update_indices()` function iterates through all trajectory buffers, ensuring that the progress index of every lane remains synchronized with the vehicle's current pose.

The lookahead distance l_a for each lane is computed dynamically based on the current vehicle velocity v_{curr} and a lane-specific lookahead ratio r :

$$l_a = \min\left(\max\left(l_{\min}, \frac{v_{\text{curr}} \cdot l_{\max}}{r}\right), l_{\max}\right), \quad (2)$$

where $l_{\min}$ and $l_{\max}$ denote the minimum and maximum lookahead bounds, respectively. Because all indices are updated in parallel, transitions between trajectories are computationally instantaneous. Upon receiving a tactical lane-switch command, the controller transforms the pre-synchronized target point into the vehicle's `base_link` frame to compute the raw steering angle δ_{raw} :

$$\delta_{\text{raw}} = \text{atan}\left(\frac{K_p \cdot 2L \sin(\alpha)}{l_a}\right). \quad (3)$$

To address the localization covariance collapse observed during high-speed transitions, a **slew-rate limiter** is applied to the control output. This constrains the change in steering angle between consecutive time steps:

$$|\delta_t - \delta_{t-1}| \leq \Delta\delta_{\max}, \quad (4)$$

where $\Delta\delta_{\max}$ denotes the maximum allowable angular displacement per control cycle. This smoothing prevents impulsive lateral motion that can cause particle filter divergence and enables stable transitions between racelines during aggressive defensive maneuvers.

D. Strategic Lane Selection Logic (Tactical layer)

The Raceline Manager ROS2 node implements a finite-state machine (FSM) that receives the detected bounding box center coordinates and estimated opponent distance from the perception layer and publishes the active raceline index (0: outer, 1: middle, 2: inner) to the control layer.

The FSM maintains three distinct behavioral states corresponding to the current raceline in use. From the middle state (default), the system switches to the inner raceline if the opponent's normalized horizontal position satisfies $b_n < c_\ell - b_c$, or to the outer raceline if $b_n > c_\ell + b_c$, where $c_\ell = 0.55$ is the image center offset and $b_c = 0.10$ is the position buffer. If the opponent is detected within the buffer region, the current

raceline remains optimal and no transition occurs. The buffer is intentionally biased toward the right side of the image to account for the track's persistent left-hand cornering behavior. From the outer state, transitions to the middle raceline occur only if the opponent is detected on the inner side and vice versa from inner state.

Timed safeguards prevent oscillatory raceline switching behaviors and provide robustness against sporadic detection failures. Lane changes are gated by a proximity threshold ($d < 2.0$ m), ensuring tactical interventions occur only when necessary. Additionally if no opponent is detected for approximately 2 s, the system defaults to its optimal middle raceline.

IV. EXPERIMENTAL SETUP AND RESULTS

The proposed framework was evaluated on a real FITENTH vehicle equipped with an NVIDIA Jetson Orin Nano, a ZED2 RGB-D camera, and a Hokuyo LiDAR. Experiments were conducted on a closed-track environment over a four-week period. The initial phase focused on training the YOLOv11 detection model and tuning depth-estimation thresholds. The second phase involved generating racelines and high-speed validation of the multi-lane Pure Pursuit controller to ensure steering stability during lane transitions. Final testing pitted the ego-vehicle against a human-controlled opponent attempting various overtaking maneuvers on both straightaways and corners

A. Results and Observations

Performance was measured by the system's ability to detect a trailing opponent and prevent successful overtaking attempts. In successful trials, the tactical manager identified the approaching vehicle and transitioned to a defensive race-line (Inner or Outer) before the opponent reached a passing position. The ego vehicle maintained stable tracking of the selected raceline while preserving a safe margin from track boundaries.

B. Challenges and Future Work

Two primary limitations were identified. First, the rear-facing camera occasionally lost track of the opponent during side-by-side passing, indicating the need for multi-camera or 360° perception coverage.

Second, aggressive maneuvers occasionally introduce transient localization errors in the Particle filter. While typically recovered within one control cycle, delayed recovery can lead to trajectory boundary violations during rapid multi-lane switches, particularly when transitioning to outer racelines during corner exit, potentially compromising vehicle safety.

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